REFINED EXPERT RF DESIGN: TRIANGULAR STAGGERED LAYOUT

1. VISUAL DESIGN LAYOUT (STAGGERED TRIANGLE)

The staggered triangular layout ensures that the 'Weak Spot' of any two APs is covered directly by a third AP. This eliminates coverage holes.

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ROW 1:  [AP-01]-----------(15m)-----------[AP-02]-----------(15m)-----------[AP-03]
           \             /           \             /
            \   (13m)   /             \   (13m)   /
             \         /               \         /
ROW 2:       [AP-04]-----------(15m)-----------[AP-05]
             /         \               /         \
            /   (13m)   \             /   (13m)   \
           /             \           /             \
ROW 3:  [AP-06]-----------(15m)-----------[AP-07]-----------(15m)-----------[AP-08]

*Legend: (15m) = Horizontal Spacing | (13m) = Staggered Vertical Spacing (0.866 * 15m)
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2. DISTANCE & SPACING FORMULAS

To achieve expert-level mobilization for Voice, the 'Cell Overlap' must be at least 20%.
- For 5GHz (Standard Enterprise): Distance between APs should be 15 meters (approx 50 feet).
- For High Ceiling / Industrial: Distance should be reduced to 12 meters (approx 40 feet).

Why this distance?
In an indoor environment with standard walls, a Wi-Fi signal drops to -65 dBm (the threshold for voice) at approximately 7-8 meters from the AP. By placing APs 15 meters apart, the 'overlap' occurs at the 7.5-meter mark, ensuring continuous -65 dBm coverage throughout the floor.

3. CHANNEL REUSE STRATEGY

In a triangular grid, every adjacent AP must use a non-overlapping channel to prevent Co-Channel Interference (CCI).
- 2.4 GHz Plan: AP-01 (Ch 1), AP-02 (Ch 6), AP-04 (Ch 11).
- 5 GHz Plan: AP-01 (Ch 36), AP-02 (Ch 44), AP-04 (Ch 52).
This creates a 'Channel Cluster' where no two APs in a triangle share the same frequency.

4. SUMMARY FOR ARCHITECTS

1. Design for 5GHz: Always plan based on 5GHz propagation; 2.4GHz will naturally follow but must be power-tuned (reduced) to match 5GHz cells.
2. Fast Roaming: Enable 802.11r (Fast Transition) to allow the 'Mobilization' required for Voice.
3. Triangular Spacing: Use the 15-meter staggered rule to ensure N+1 redundancy.